Final Project Guidelines

CS3270 Intelligent Systems, Spring 2026

Total Final Weight (5% + 15% =20%)

Grading (total 20%):

- **Phase 1:** Final Group Project Proposal Presentation (Total points 100 and final weight 5%) [March 25, 2026, Wednesday 10:30 AM (In person)]
- **Phase 2:** Final Group Project Demo and Submission (Total points 100 and final weight 15%) [April 29, 2026, Wednesday 10:30 AM (In person)]

Project Goal: The goal of this group project is to apply students' understanding of Intelligent Systems and their implementation processes to develop machine learning (ML) or other relevant models that solve real-life artificial intelligence (AI) application problems. Students will select and apply one or more approaches such as AI, ML, data science, heuristic programming, or agent-based programming based on the specific requirements of their chosen problem, while following best practices. If interested, students may also use sensors to collect data by setting up a testbed and using the collected data as input to their models. They are expected to utilize open-source and royalty-free data and code (if needed), while practicing and recognizing professional and ethical responsibilities related to the design, implementation, and use of intelligent systems. Students will work effectively as team members to apply fundamental AI/ML concepts to real-world intelligent systems problems. Through two in-person presentations and a written report, students will develop the ability to communicate technical results clearly and accurately to both technical and non-technical audiences, in both written and oral formats.

Phase 1: Final Group Project Proposal Presentation (March 25, 2026, Wednesday 10:30 AM (In person) [5% in final grade and 100 points]: Presentation order group 9 to group 1

1. Understanding your business model (dataset description and what you want to do with that dataset) [20 points]
2. What kind of AI/ML/ intelligent-system mechanisms do you want to use and why? [20 points]
3. What is your expected output? [20 points]
4. Project plan and task distribution among team members [20 points]
5. Upload your slides in Moodle [after presentation] [20 points]

Deliverable:

1. Presentation slides [Every team member needs to upload it to Moodle individually]

Submission procedure: Each group member will upload your presentation to Moodle before **Deadline**. File name: CS3270_Group#_YourNAMEs_YourIDs.ppt/pdf

Phase 2: Final Group Project Demo and Submission (April 29, 2026, Wednesday 10:30 AM (In person)) [15% in final grade and 100 points]: Demo order group 1 to group 9

1. How the intelligent-system life cycle reflects in your project development [20 points]

2. Describe your model [10 points]
3. Output and results [10 points]
4. Your findings and discussion [10 points]
5. How you executed your project plan [10 points]
6. Code, and demo presentation upload in Moodle [20 points]
7. Final report upload to Moodle [10 points]
8. Code upload to Colab/GitHub Repo and share your code link with instructor [10 point]

Deliverable:

1. Presentation slides/demo, code, report, code link. [Every team member needs to upload it to Moodle individually]

Submission procedure: Each group member will upload your presentation, code, and report in doc/pdf as a zip folder to Moodle before **Deadline**.

Zip file name: CS3270_Group#_YourNAMEs_YourIDs.zip

If you have any questions or discussion, please utilize the classroom meeting, my office hours, or online meeting by appointment.

Outcomes:

- Understand the Intelligent Systems and their implementation process.
- Understand the role of artificial intelligence (AI), machine learning (ML), data science, heuristic programming, and agent-based programming for Intelligent Systems.
- Apply heuristic and optimal search methods to solve a problem.
- Use ML methods to solve a problem by following best practices.
- Experiential learning of Intelligent Systems based on AI/ML, sensors, Raspberry PI, and corresponding software.
- Recognize professional and ethical responsibilities in relation to the design, implementation, and use of intelligent systems.
- Communicate technical results to non-technical audiences clearly and correctly, in writing and verbally.
- Effectively function as a member of a team to apply fundamental concepts of AI/ML to solve real-world intelligent systems problems

Here is the list of sample projects, you can get an idea about your expected project. You can choose any domain from here and you can propose your own.

1. https://www.kaggle.com/datasets/emmarex/plantdisease
2. https://www.kaggle.com/datasets/dhamur/cotton-plant-disease
3. https://www.kaggle.com/datasets/mgmitesh/plant-disease-detection-dataset
4. https://www.kaggle.com/datasets/vasukipatel/face-recognition-dataset
5. https://universe.roboflow.com/pklot-parking-detection/empty-car-parking-spot-detetection
6. https://www.kaggle.com/datasets/trainingdatapro/parking-space-detection-dataset

7. https://www.youtube.com/watch?v=aE_J7B-O4VQ&ab_channel=SriTuHobby
8. https://www.youtube.com/watch?v=etVX0XkBX4o&ab_channel=RobotLk
9. https://www.kaggle.com/datasets/mostafaabla/garbage-classification
10. https://www.kaggle.com/datasets/sumn2u/garbage-classification-v2
11. https://www.kaggle.com/datasets/moltean/fruits
12. https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset
13. https://www.kaggle.com/datasets/mfarhaannazirkhan/heart-dataset
14. https://www.kaggle.com/datasets/dheerov/stress-detection-dataset
15. https://automationroboticsarduino.com/products/a-r-nano-uno-mega-colossus-starter-kit-for-arduino
16. https://www.youtube.com/watch?v=vrk8wm5rBNE&ab_channel=AnotherMaker
17. https://projecthub.arduino.cc/lee_curiosity/a-beginners-guide-controlling-an-arduino-car-with-a-remote-5d4418

Group 1:

Moria El Akaya

Donovan Bonner

Group 2:

Brailey Sharpe

Matthew Shirley

Colby Daniel

Group 3:

Kelis Gates

Azizah Blair

Group 4:

Will Lynch

Evan Olah

Joseph Stiltner

Group 5:

Kellen Andrews

JJ Araim

Group 6:

Gretchen Kuba

Simisola Omotoso

Group 7:

Samuel Amihere

Mickel Murenzi

Abdul Azeez Dauda

Group 8:

Ethan Gray

Hector Trabado

Group 9:

Samantha Cook

Everett Dewberry